

Exam 3 Review Assessment

1

Which of the following reactions would have the largest equilibrium constant K_{eq} ?

- A. $\text{ClCH}_2\text{CH}_2\text{COOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{ClCH}_2\text{CH}_2\text{COO}^-(aq)$
- B. $\text{ClCH}_2\text{COOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{ClCH}_2\text{COO}^-(aq)$
- C. $\text{CH}_3\text{COOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{CH}_3\text{COO}^-(aq)$
- D. $\text{Cl}_2\text{CHCOOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{Cl}_2\text{CHCOO}^-(aq)$
- E. $\text{Cl}_3\text{CCOOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{Cl}_3\text{CCOO}^-(aq)$

oxyacids (like our strong acid HClO_4)

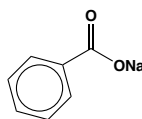
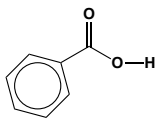
↳ Rules for oxyacids:

1. acid strength increases w/ # of oxygens
2. acid strength increase w/ # of electroneg atoms + the stronger electroneg atoms
↳ these polarize the O-H bond, making it more readily deprotonated

due to these rules, compare # of oxygens (all the same)
+ compare # of electroneg atoms → using that E. would be the strongest

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The pK_a of benzoic acid (below, left) is 4.19. If Julia has 200 mL of a solution that is 0.2 M in benzoic acid and 0.1 M in sodium benzoate, what will the pH of the solution be after the addition of 10 mL of 0.3 M calcium hydroxide at 298 K?



$$(200 \text{ mL})(0.2 \text{ M}) = 40 \text{ mmol HA}$$

$$(200 \text{ mL})(0.1 \text{ M}) = 20 \text{ mmol A}^-$$



$$[\text{OH}^-] = 0.6 \text{ M}, (10 \text{ mL})(0.6 \text{ M}) = 6 \text{ mmol}$$

- A. 4.07
- B. 4.30
- C. 3.96
- D. 3.89
- E. 3.98

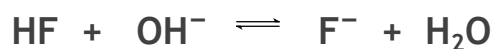
	OH^-	HA	H_2O	A^-
I.	6	40		20
C.	-6	-6		+6
F.	0	34		26

$$\text{pH} = 4.19 + \log(26/34)$$

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- What is the equilibrium constant equal to for the following reaction?



- A. K_a
- B. K_w/K_a
- C. K_w/K_b
- D. $1/K_a$
- E. K_a/K_w

Reverse reaction is base hydrolysis of fluoride ion.
Base hydrolysis has a K_b .
The reverse reaction has $1/K_b$.
Since $K_b = K_w/K_a$, then $1/K_b = K_a/K_w$.

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The acid-dissociation constants of ascorbic acid ($\text{H}_2\text{C}_6\text{H}_6\text{O}_6$) are as follows at 25.0°C . Calculate the pH of a 2.5 M aqueous solution of ascorbic acid.

$$K_{a1} = 8.0 \times 10^{-5}$$

$$K_{a2} = 1.6 \times 10^{-12}$$

- A. 2.51
- B. 0.40
- C. 1.85
- D. 0.88
- E. 0.13

Use K_{a1} only. Write acid hydrolysis reaction and do an ICE table. Sub the E line of the table into the K_{a1} expression.

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Which one of the following aqueous solutions would have the lowest pH?

- A. CH_3COONa
- B. MoCl_2
- C. $\text{Ca}(\text{OH})_2$
- D. BaCl_2
- E. K_2S

Dissociate all of these salts. Charged transition metal ions are Lewis acids and can accept lone pairs of electrons from ligands.

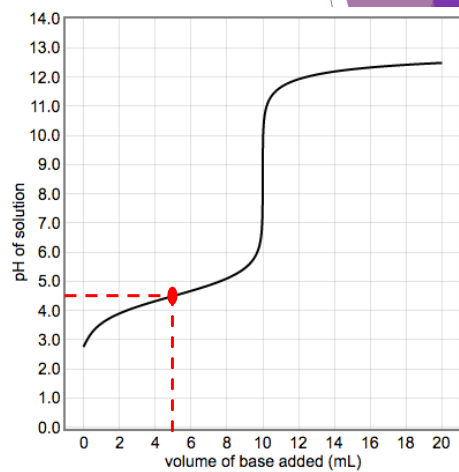
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What is the approximate K_a of the substance being titrated in the curve below?

- A. 3.2×10^{-5}
- B. 2.0×10^{-7}
- C. $K_a \gg 1$
- D. 1.0×10^{-9}
- E. 1.2×10^{-2}



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What is the pH of a 0.5 M LiOBr solution?

K_a for HOBr = 2.3×10^{-9}

	$\text{OBr}^- + \text{H}_2\text{O} \rightarrow \text{HOBr} + \text{OH}^-$		
I.	0.5 M	0	0
C.	-x	+x	+x
E.	0.5 - x	x	x

$$K_b = K_w/K_a = 4.348 \times 10^{-6} = x^2/0.5$$

$$\text{pOH} = 2.83$$

- A. **11.2**
- B. 2.83
- C. 5.36
- D. 8.64
- E. 1.18

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